PROGRAMMING WITH PYTHON · SEMESTER 5

AI Blog Generator

Turning YouTube videos into blog articles with Django

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INTRODUCTION

A video in, an article out

THE PROBLEM

A video can’t be skimmed, searched or quoted like text. Turning it into an article by hand takes hours.

THE SOLUTION

A Django web app that turns any YouTube link into a saved, readable blog article.

OBJECTIVES

- Build a working Django web app
- Secure signup, login and logout
- Turn a YouTube link into an article
- Match the tone to the video title
- Save each article in PostgreSQL
- Show posts only to their owner

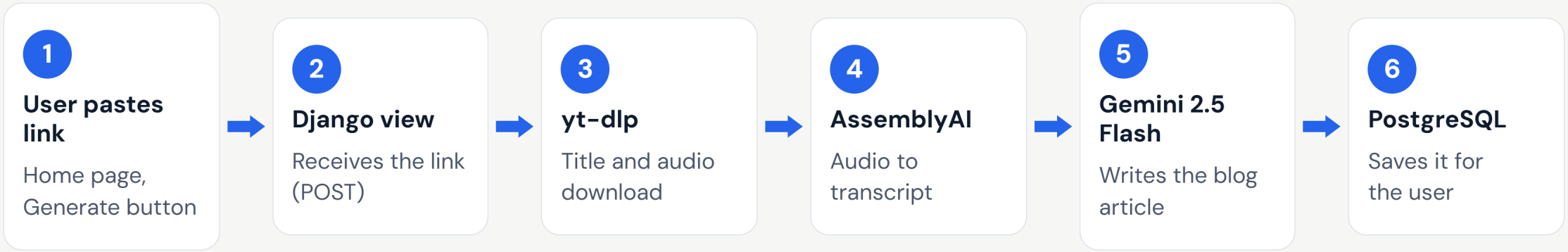
TECHNOLOGY STACK

Python at the center, three tools around it

Layer	Technology	Role in the project
Language	Python 3.13	Backend logic and all integrations
Web framework	Django	Routing, views, templates, authentication, ORM
Database	PostgreSQL	Stores users and generated blog posts
Audio download	yt-dlp	Fetches the title and downloads the audio
Speech-to-text	AssemblyAI	Converts the audio into a transcript
Text generation	Google Gemini 2.5 Flash	Writes the blog article from the transcript
Frontend	HTML, Tailwind CSS, JavaScript	Pages, styling and the call to the backend
Tools	VS Code, uv, pgAdmin, Git, GitHub	Development, packages, database, version control

HOW IT WORKS

One request runs the whole pipeline



BEHIND THE SCENES

Steps 3 to 5 use third-party tools (yt-dlp, AssemblyAI and Gemini). Steps 2 and 6 are handled by Django itself, and the saved article is returned to the page.

MODULES

Three modules in one Django app

1. Authentication

- Signup checks both passwords match
- Login starts a session
- Logout ends it
- Home page needs a login

2. Blog generation

- Reads the title with yt-dlp
- Downloads the audio
- AssemblyAI makes the transcript
- Gemini writes the article
- Saved as a BlogPost

3. Saved posts

- List of your own articles
- Details page for each article
- Other users' posts redirect home

Pages: / · /login · /signup · /generate-blog · /blog-list · /blog-details/<id>/

DATABASE DESIGN

One user has many blog posts

User

Django built-in table

- username
- email
- password (stored hashed)

one user



many posts

BlogPost

Stores every generated article

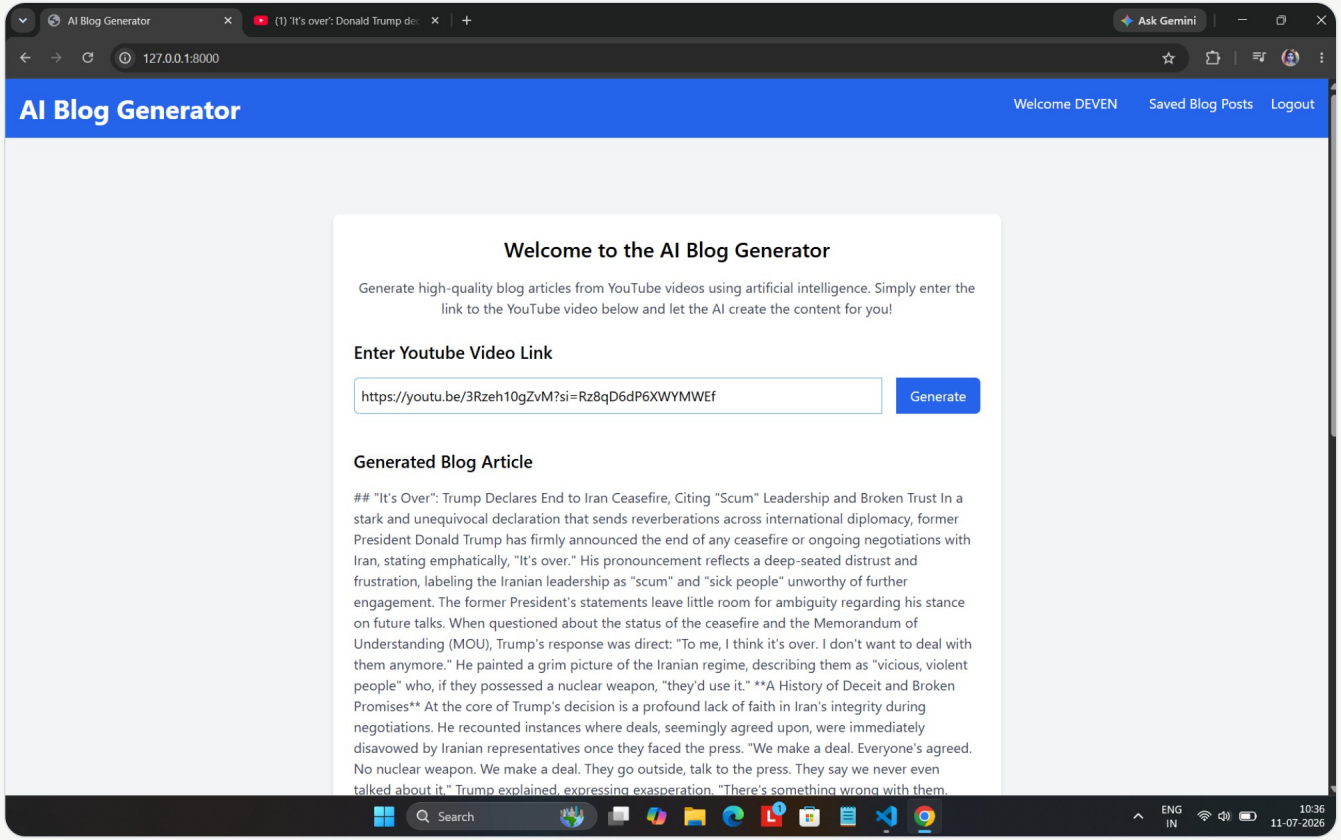
- user (foreign key to User)
- youtube_title
- youtube_link
- generated_content

Tables are created with Django migrations. PostgreSQL is connected through the DATABASES setting in settings.py.

APPLICATION SCREENS

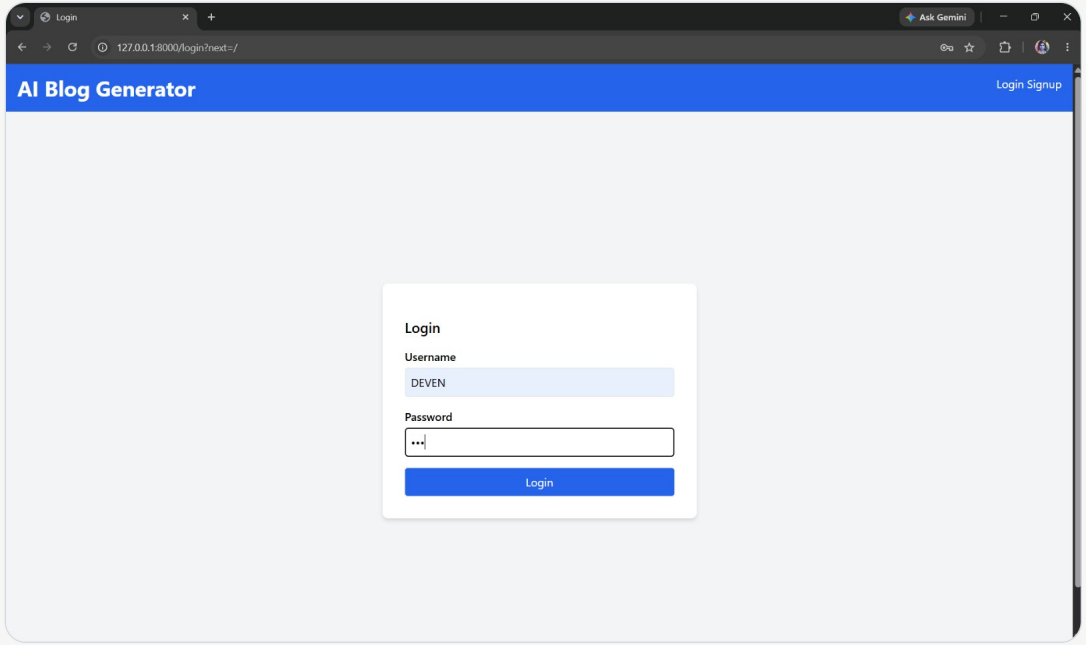
Home page: link in, article out

1. Paste a YouTube link
2. Press Generate
3. The article appears below
4. It is saved automatically

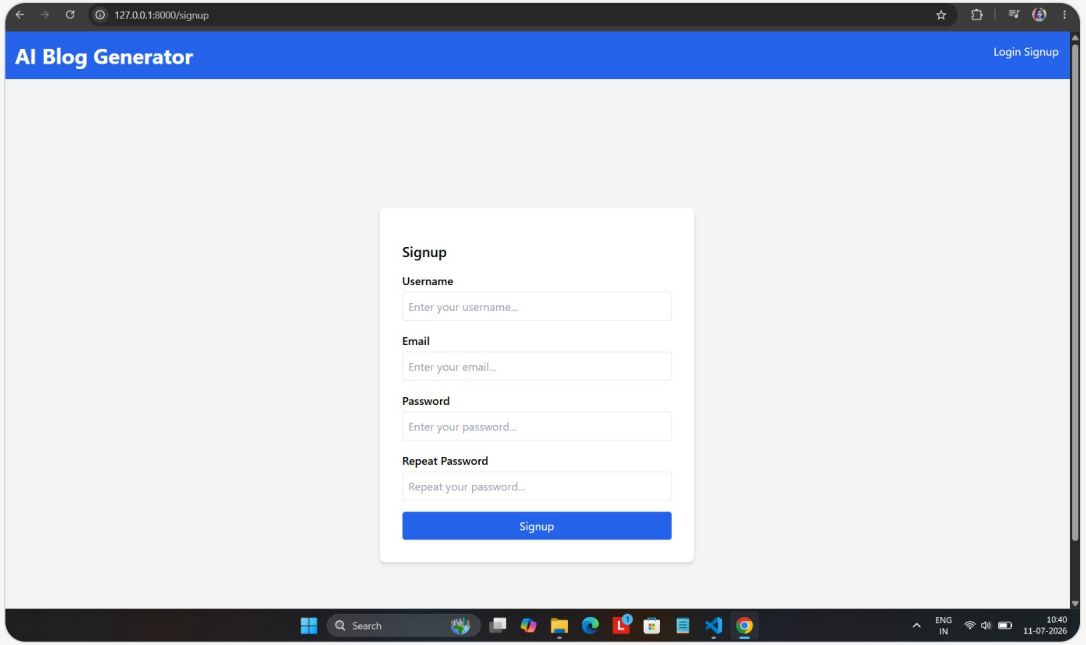


APPLICATION SCREENS

Login and signup



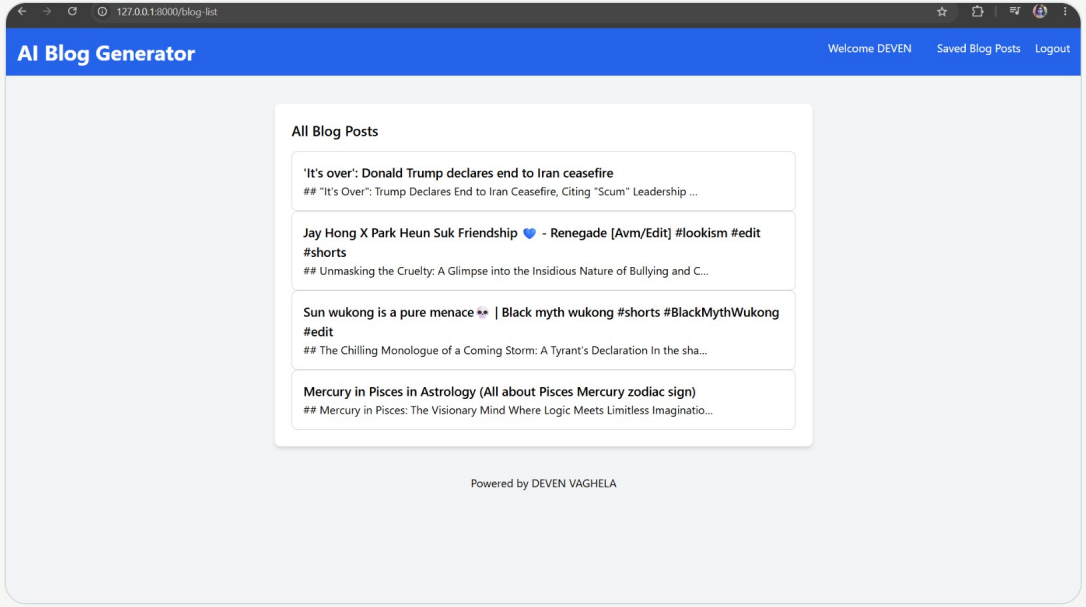
Login page



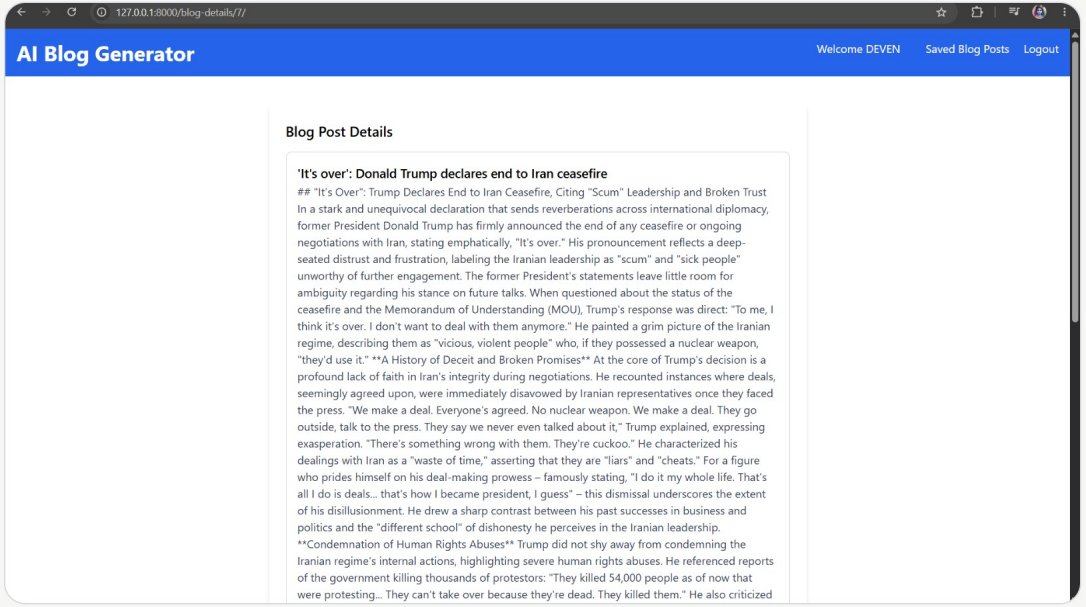
Signup page

APPLICATION SCREENS

Saved posts and blog details



All saved blog posts



Blog details with title and link

CHALLENGES

Problems met and how they were solved

Challenge	Solution
YouTube blocked the old download library with a bot check	Switched to yt-dlp, which is updated often
Renaming the audio file to .mp3 failed on Windows	Removed the rename; AssemblyAI accepts the downloaded format
API keys and the database password must stay off GitHub	Placeholders in the code and a .gitignore for private files
A failing service could crash the request	Error handling around the title and transcript steps

LIMITATIONS AND FUTURE SCOPE

What is missing and what comes next

CURRENT LIMITATIONS

- Depends on YouTube, AssemblyAI and Gemini
- Long videos take longer to process
- Markdown symbols (## and **) show as plain text
- API keys are written in the code
- A failed Gemini call saves its error message as the article

FUTURE SCOPE

- Read keys from environment variables
- Show Markdown as real headings and bold text
- Edit and delete saved posts
- Export articles as PDF or Markdown
- Choose the output language and style
- CSRF handling and Docker deployment

CONCLUSION

A YouTube link goes in, a saved blog article comes out

The project combines Django authentication, PostgreSQL and three outside tools in one working application. When YouTube blocked the first download library, switching to yt-dlp kept it running.

Thank you

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Source code

github.com/devenvaghela57-cloud/ai-blog-generator